LendInsight

Analytics & Business Intelligence Report

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Executive Summary

LendInsight is an Analytics & Business Intelligence platform developed to analyze customer loan data, identify high-risk applicants using Machine Learning, segment customers using K-Means clustering, and generate actionable insights for financial institutions. This report summarizes customer demographics, predictive analytics, and business recommendations that support data-driven lending decisions.

Key Performance Indicators

KPI	Value
Total Customers	252,000
Average Income	■4,997,116.67
High Risk Customers	30,996
States Covered	29

Machine Learning Performance

Metric	Score
Accuracy	89.91%
Precision	60.19%
Recall	53.99%
F1 Score	56.93%

Feature Importance

Feature	Importance
income	0.4701
age	0.2537
experience	0.0971
current_job_yrs	0.0906
current_house_yrs	0.0885

Top States

State	Customers
Uttar_Pradesh	28,400
Maharashtra	25,562
Andhra_Pradesh	25,297
West_Bengal	23,483
Bihar	19,780
Tamil_Nadu	16,537
Madhya_Pradesh	14,122
Karnataka	11,855
Gujarat	11,408
Rajasthan	9,174

Top Professions

Profession	Customers
Physician	5,957
Statistician	5,806
Web_designer	5,397
Psychologist	5,390
Computer_hardware_engineer	5,372
Drafter	5,359
Magistrate	5,357
Fashion_Designer	5,304
Air_traffic_controller	5,281
Comedian	5,259

Customer Segmentation

Customers are segmented into Premium, Regular and Budget groups using K-Means Clustering based on income, age, experience, current job years, and current house years. These segments help financial institutions personalize loan offerings and improve customer relationship management.

Risk Analysis

The Random Forest classifier predicts customer loan risk using demographic and financial attributes. High-risk applicants can be prioritized for manual review while low-risk customers can benefit from faster loan approval, improving operational efficiency.

Business Recommendations

• Prioritize manual verification for customers predicted as High Risk.

• Design personalized loan products using customer segmentation.

• Monitor state-wise customer trends to optimize lending strategies.

• Regularly retrain the Machine Learning model using fresh customer data.

• Use feature importance analysis to improve underwriting policies.

• Build executive dashboards for continuous portfolio monitoring.

Conclusion

LendInsight demonstrates how Business Intelligence, SQL Analytics and Machine Learning can be integrated into a single platform for modern financial institutions. The solution enables customer segmentation, predictive loan risk assessment, analytical reporting and executive-level decision support through an interactive dashboard and automated PDF reports.